

AI-Based Classroom Photo Attendance Management System Using Face Recognition

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Abstract—Traditional Radio Frequency Identification (RFID)-based attendance systems in educational institutions face significant limitations including card dependency, proxy attendance, time consumption during large lectures, and lack of real-time verification. This paper proposes an AI-based Classroom Photo Attendance Management System that leverages facial recognition technology to automate attendance tracking. The system enables instructors to capture a single photograph of the classroom using any device, which is then processed through a deep learning pipeline for face detection and recognition against a pre-registered student database. The proposed solution addresses critical challenges including privacy concerns, recognition accuracy under varying conditions, late arrivals, proxy prevention, cross-platform compatibility, and dataset limitations. Built with Python, Flask, OpenCV, FaceNet/dlib, and React.js/Flutter, the system aims to reduce administrative overhead, eliminate proxy attendance, and provide faculty with an intuitive dashboard for attendance review and export. This paper presents the system architecture, technical challenges, proposed solutions, and expected outcomes, representing the planning phase of our senior design project.

Index Terms—Face Recognition, Attendance Management, Deep Learning, Biometric Authentication, Classroom Automation, Computer Vision, RFID Alternative

I. INTRODUCTION

Educational institutions worldwide are increasingly adopting automated systems to streamline administrative processes. Attendance tracking, a fundamental yet time-consuming task, has traditionally relied on manual roll calls or RFID card systems. At North South University (NSU), the existing RFID-based system requires students to scan their cards upon entering the classroom. While this approach reduces manual effort, it introduces several limitations: students who forget their cards cannot mark attendance despite being physically present; proxy attendance remains possible through card sharing; queues form during large lectures, consuming valuable instructional time; and the system cannot verify whether the card owner is actually present in the classroom.

The emergence of computer vision and deep learning technologies has opened new possibilities for automating attendance through facial recognition. By processing classroom photographs captured by instructors, such systems can identify enrolled students without requiring physical tokens or manual roll calls. This approach promises to reduce administrative burden, eliminate proxy attendance, and provide real-time verification.

This paper presents the design and planning of an AI-Based Classroom Photo Attendance Management System that uses facial recognition to automate attendance tracking in university classrooms. The proposed system enables instructors to capture a single classroom photograph using any device, which is then processed by an AI model to detect faces and match them against a pre-registered student database. The system addresses critical challenges including privacy preservation, accuracy optimization, late-arrival handling, proxy prevention, cross-platform compatibility, and dataset collection strategies.

II. EXISTING SYSTEM AND ITS LIMITATIONS

A. Current RFID-Based System at NSU

North South University currently employs an RFID-based attendance system. Upon entering the classroom, each student scans their RFID card on a wall-mounted sensor, and attendance is automatically recorded in the university's central system (RDS). While this system reduces manual roll-call efforts, it presents several fundamental limitations.

B. Limitations of RFID Systems

- **Card Dependency:** Students who forget or lose their RFID cards cannot mark attendance, even when physically present in the classroom. This creates frustration and unfair penalties.
- **Proxy Punching:** Classmates can easily scan cards on someone else's behalf, undermining the integrity of attendance records and academic honesty.

- **Time-Consuming:** During large lectures, queues form at the sensor, consuming valuable lecture time and disrupting the learning flow.
- **Hardware Maintenance:** RFID readers require installation, maintenance, and periodic replacement due to wear and technical issues, incurring ongoing costs.
- **No Real-Time Verification:** The system cannot confirm whether the card owner is actually inside the classroom, enabling fraudulent attendance marking.

III. PROPOSED SOLUTION

The proposed AI-Based Classroom Photo Attendance Management System replaces the physical RFID card with facial recognition technology. The instructor captures a single photograph of the entire classroom from the front using any device, including phones, tablets, or laptop cameras. This image is sent to our application where an AI model detects every face and matches it against a pre-registered student database stored in the backend.

A. System Workflow

- 1) **Photo Capture:** The instructor takes a photo of the classroom using any device capable of image capture.
- 2) **Face Detection:** The software detects all faces present in the captured image using computer vision algorithms.
- 3) **Face Matching:** Each detected face is compared against the student database using facial recognition models.
- 4) **Attendance Recording:** Matched students are automatically marked present, while unmatched students are marked absent after the configured time window closes.

B. Key Advantages Over RFID Systems

- No physical tokens required
- Elimination of proxy attendance through biometric verification
- No queues or wait times for students
- Real-time verification of physical presence
- Reduced hardware maintenance overhead

IV. TECHNICAL CHALLENGES AND SOLUTIONS

Building a reliable facial-recognition-based attendance system requires addressing several technical and ethical challenges. This section outlines the critical "W" and "H" questions and our proposed solutions.

A. Privacy Concerns

Challenge: Student facial data constitutes sensitive biometric information that requires robust protection.

Solution: All facial embeddings will be encrypted both at rest and in transit. Database access will be restricted via role-based permissions, and the system will be designed for on-campus deployment, ensuring data never leaves NSU servers. Additionally, we will implement data retention policies that automatically purge unnecessary data.

B. Recognition Accuracy

Challenge: Poor lighting conditions, angled faces, and partial occlusion can significantly reduce recognition accuracy in real classroom environments.

Solution: We will train our model on diverse datasets that include variations in lighting, pose, and facial expressions. Image pre-processing steps including contrast enhancement, alignment, and noise reduction will be applied. Detection thresholds will be fine-tuned through iterative testing in actual NSU classrooms to achieve optimal performance.

C. Late Arrival Handling

Challenge: A rigid cutoff time could unfairly penalize students who arrive late due to legitimate reasons.

Solution: The system will include a configurable grace period, defaulting to 20 minutes after class start time. During this window, late-arriving students can still be recognized and marked present. After the grace period expires, unmatched students are automatically marked absent.

D. Proxy Attendance Prevention

Challenge: While faces are more difficult to fake than RFID cards, twins or look-alikes could cause false positives.

Solution: We will employ deep metric learning models that map faces to high-dimensional embedding spaces, significantly reducing misidentifications among similar-looking individuals. The system will maintain a confidence threshold that can be adjusted to balance between false positives and false negatives.

E. Training Data Acquisition

Challenge: Pre-built datasets may not reflect NSU's student demographics, potentially impacting recognition accuracy.

Solution: We will combine public datasets like Labeled Faces in the Wild (LFW) with a locally collected dataset of consenting NSU students, captured under real classroom lighting conditions. Students will register their profiles during admission, providing 5-10 facial images from different angles and expressions.

F. Cross-Platform Compatibility

Challenge: Instructors use various devices including PCs, Macs, and smartphones.

Solution: We will build a responsive cross-platform frontend using React.js (web) and Flutter (mobile), along with a cloud-backed REST API built with Flask. This architecture ensures consistent performance regardless of the device used to capture or review attendance.

V. TECHNOLOGY STACK

A. Backend & AI Layer

- **Python with Flask:** REST API development and server logic
- **OpenCV:** Face detection and image preprocessing
- **FaceNet/dlib:** Facial embedding and recognition
- **MySQL/PostgreSQL:** Student and attendance database management

B. Frontend & Development Tools

- **React.js/Flutter:** Cross-platform UI for PC, Mac, and mobile devices
- **Firebase/Docker:** Deployment and containerized hosting
- **Git & GitHub:** Version control and team collaboration
- **Figma:** UI/UX prototyping and design
- **Postman:** API testing and documentation

VI. EXPECTED OUTCOMES

A. Automated Attendance Recording

The system will process classroom photographs into reliable attendance logs with timestamps and confidence metrics, eliminating manual roll-call efforts.

B. Time Savings

Significant reduction in class time lost to manual roll-calls and administrative overhead, allowing instructors to focus on teaching.

C. Accuracy and Fraud Reduction

Lower incidence of proxy attendance through robust identity verification and configurable confidence thresholds.

D. Usable Dashboard and Reports

Faculty will have access to an intuitive dashboard for reviewing, correcting, and exporting attendance data for record-keeping and analytics.

VII. CONCLUSION AND FUTURE SCOPE

This paper presents the design and planning of an AI-Based Classroom Photo Attendance Management System using facial recognition technology. By addressing the limitations of traditional RFID-based systems, our proposed solution offers a more reliable, efficient, and fraud-resistant approach to attendance tracking in educational settings.

A. Future Scope

- **Version 2:** Hybrid RFID + face verification combining convenience with stronger anti-proxy checks
- **Version 3:** CCTV-based continuous monitoring and real-time alerts for absenteeism patterns
- **Mobile App:** Student self-check and push notifications for attendance status
- **Advanced Analytics:** Longitudinal attendance trends, at-risk student detection, and integration with Learning Management Systems (LMS)